

MULTIDRIVE DUAL-MOTOR MODULE

Model: DMM-MD24V-REV3 | Installation & Specification Sheet

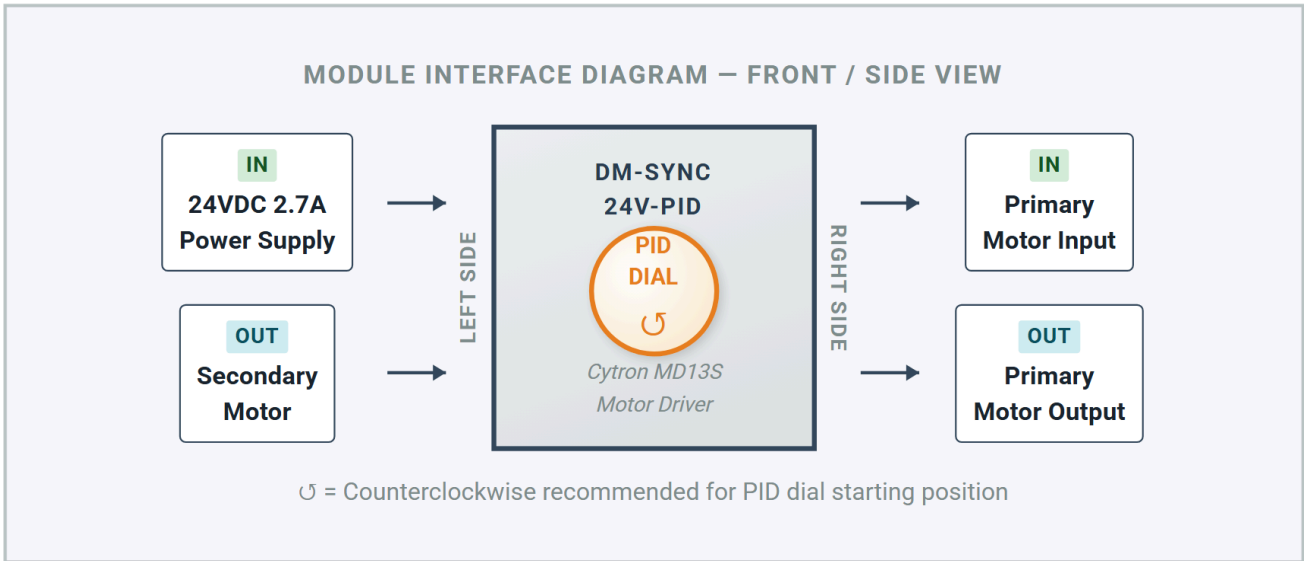
Product Overview

The DMM-MD24V-REV3 (Dual Motor Module) is a control module designed for the Autoslide MultiDrive automatic sliding door system, employing a dual-motor configuration on large sliding doors requiring heavy opening and closing force. This unit ensures smooth, synchronized operation by actively monitoring the current draw of the primary motor and adjusting the secondary motor in real-time. The module is compliant with the existing sliding rail component system in the MultiDrive, and should be slid directly adjacent to the main controller and motor.

Technical Specifications

Compatible Motor Type	Autoslide LLC MultiDrive 24VDC Brushed Motor
Power Supply Input	24VDC, 2.7A (Regulated)
Secondary Motor Driver	Cytron MD13S (Integrated)
Control Methodology	Active PID Current Equalization Loop (adjustable via dial)
Circuit Protection	Polyfuse (Overcurrent), TVS Diode (Transient Voltage Suppression)
Mounting Interface	Standard railing system slide-mount

Interface & Port Layout



Installation Instructions

Necessary Parts

- 1x Dual Motor Module
- 1x MultiDrive Motor (to be used as secondary motor)
- 1x MultiDrive Power Supply (to be used as secondary power supply)
- MultiDrive Motor Extension Cabling
- 1x Complete MultiDrive Kit (includes primary motor, controller, primary power supply)

Part 1 - Physical Setup

1. **Pre-Assembly:** Prior to mounting, pre-assemble the MultiDrive system per its standard instructions. In place of a belt tensioner on one side of the door, install the secondary motor and secondary power supply. Both power supplies should be located near the MultiDrive controller. Position the Dual Motor Module immediately next to the controller and primary motor for optimal cable routing.
2. **Motor Mounting:** Tighten one motor securely in place, and leave the other motor very slightly loose so it can be adjusted during belt tensioning.
3. **Belt Routing:** After mounting the system onto the door, feed the MultiDrive belt through both motors. Pull the belt tightly (while ensuring both motors remain in place), then cut and connect the belt ends per the standard MultiDrive instructions.
4. **Belt Tensioning:** Using a prying tool or ratchet strap, manually pull the loose motor as far from the fixed motor as possible to tension the belt. If necessary, have a helper support the middle of the belt to minimize slack. The belt should be as tight as possible to prevent slipping before tightening the second motor fully in place.

Part 2 - Cable and Module Setup

1. **Primary Motor IN (Right Side):** Plug the cable coming from the primary motor — the motor closest to the MultiDrive Controller — into the open Primary Motor IN port on the right side of the Dual Motor Module.
2. **Primary Motor OUT to Controller (Right Side):** Plug the motor cable coming out of the Primary Motor OUT port on the right side of the Dual Motor Module into the Motor port on the MultiDrive Controller.
3. **Secondary Motor (Left Side):** Plug the secondary motor — the motor furthest from the MultiDrive Controller — into the open Secondary Motor OUT port on the left side of the Dual Motor Module.
4. **Secondary Power Supply (Left Side):** Plug the 24VDC 2.7A secondary power supply into the open Power Supply IN port on the left side of the Dual Motor Module. Verify polarity before connecting.
5. **Power On & Programming:** Once all connections are complete, proceed with powering on and programming the MultiDrive system as normal.

Calibration Warning: PID Dial Adjustment

The PID tuning dial controls the responsiveness of the secondary motor relative to the primary motor.

Recommendation: Start with the dial turned fully Counter-Clockwise (Low Responsiveness). Gradually increase (Clockwise) only if the secondary motor lags significantly behind the primary motor. Over-tuning may cause oscillation or "jittering" of the door.